STEM Principles and Practices to Design Requirements *(E1)*

This section focuses on articulating how your proposed solution is supported by STEM (science, technology, engineering, mathematics, computer science, etc.) concepts. Concepts can be laws, theories, and practices, including disciplinary core ideas (i.e., Next Generation Science Standards, Common Core State Standards) and standards for mathematical practice. Go beyond describing the engineering design process to describe the STEM concepts as they apply to your proposed solution in order to meet the design requirements. Be sure to apply the applicable STEM concepts in context, directly to the specifics of your design. Review concepts learned in your other classes and tackle new concepts as appropriate. Include any relevant calculations.

C3 indicates that you should have at least three high-priority design requirements to receive a score of 3 (and four for a score of 4, and five for a 5). Your number of design requirements and your number of high priority design requirements should agree throughout your portfolio. You should use the same five design requirements here too, and apply at least four of them to STEM concepts to achieve a score of 5 on E1. Use as many copies of this sub-template as needed.

STEM Principle for High-Priority Design Requirement #1
STEM Concept:
Describe how STEM concept applies directly to the specifics of your design: Stem applies to our design because we had to engineer the mechanical movements it made.

STEM Principle for High-Priority Design Requirement #2
STEM Concept: Math
Describe how STEM concept applies directly to the specifics of your design:
We calculated the weights and dimensions of our parts for the design.
STEM Principle for High-Priority Design Requirement #3
STEM Concept:Engineering
Describe how STEM concept applies directly to the specifics of your design:
We took the drafting part out of engineering and sketched our ideas for different designs.
STEM Principle for High-Priority Design Requirement #4
STEM Concept:Science
Describe how STEM concept applies directly to the specifics of your design:
We conducted scientific experiments to make sure our design worked properly.
STEM Principle for High-Priority Design Requirement #5
STEM Concept:Technology
Describe how STEM concept applies directly to the specifics of your design:

We used batteries and 12v motors to make our project work.

Proposed Solution to STEM Principles and Practices *(E2)*

While E1 was focused on how STEM applies to the design requirements, E2 is focused on how STEM is used to support the design solution.

For E2, use STEM concepts to justify your design decisions. It's important that the STEM Principles and Practices that are discussed are actually appropriate and useful to the problem. Use as many copies of this sub-template as needed.

STEM Principle for Design Solution #1
STEM Concept:Engineering
Describe how STEM concept applies directly to the specifics of your solution:
Engineering applies to the written design and calculations along with our final product.
STEM Principle for Design Solution #2
STEM Concept:Science
Describe how STEM concept applies directly to the specifics of your solution:
Scientific experiments helped us decide what changes needed to be made and different flaws.

Evidence Design is Reviewed (E3)

"Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

E3 is about experts reviewing what you've said about the STEM connections in E1 and E2. It's important here to talk to an appropriate field expert who knows about the STEM content of the project - not to a stakeholder. We need to know what the expert said. Document in this sub-element what the expert said about the connections you've made from your design requirements to STEM principles and practices (E1) and what they said about the ways STEM principles and practices substantiate your design solution (E2). This documentation will support you to give a corrective response in E5. Potential formats for review include: interview notes, emails, audio/video tape with transcript, etc. Use as many copies of this sub-template as needed.

Documentation #1:

- What format: Interview notes, emails, audio/video tape with transcript, etc.
- What the expert said about the connection you've made from your design requirements to STEM principles and practices: We used the engineering principles well in our design.
- What the expert said about how STEM principles and practices substantiate your design solution: We used the stem principles to the best of our ability.

Experts with Credentials *(E4)*

Document who your field experts are (names are not necessary) and what qualifies them as an expert. Use as many copies of this sub-template as needed.

Expert #1: Describe credential(s) and what makes them qualified as an expert.

Duke engineering profesor, has a degree in engineering and design.

Expert #2: Describe credential(s) and what makes them qualified as an expert.

Duke professor, has a degree in math and engineering.

Expert #3: Describe credential(s) and what makes them qualified as an expert.

Duke engineering profesor, has a degree in engineering.

Verification or Corrective Response to Design Review *(E5)*

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses. For example, "__________ told us __________, which informed our decision to change/reconsider __________ ." (or "...which verified/reinforced our decision to __________."

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response: Expert 1 said that our final product worked well and was well designed but we could have planned better.

Interview with Expert #2 informs corrective response: Expert 2 said that our final product was good in the fact that it worked but we could have made it look better.

Interview with Expert #3 informs corrective response: Expert 3 said that they really liked our overall design and process but that we could have chosen better materials.